

ORIGINAL ARTICLE

Explainable artificial intelligence for wind power forecasting model based on long short-term memory

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Received: 10 January 2025 / Accepted: 26 March 2025 / Published online: 14 May 2025

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Abstract

Expanding the world's economy leads to higher requirements for energy storage. Traditional energy resources decline at the same time that environmental contamination levels increase in the world. Wind power is the most potential energy resource supported by its status as a significant renewable energy system. Wind power generation has become a popular and exciting method among nations worldwide for generating renewable energy. High wind power generation unpredictability results in unavoidable errors throughout the wind power prediction process, creating substantial difficulties in the optimal management of power systems. Wind power prediction errors remain unavoidable, but appropriate wind power uncertainty models help power system operators reduce their adverse impact on operational decision-making performance. In this paper, developed an appropriate machine learning model that efficiently forecasted wind power data through time series analysis. The long Short-Term Memory (LSTM), Gated Reference Unit (GRU), and Autoregressive Integrated Moving Average (ARIMA) are the machine algorithms used in this investigation. This paper proposed an X-double LSTM, which integrates explainable artificial intelligence (XAI) and long short-term memory (LSTM). The XAI-Shapley Additive Explanations (SHAP) model is modified to pinpoint the crucial elements affecting the power generation forecasting model's accuracy in cutting-edge solar systems. Nine metrics are used to assess the efficacy of the proposed X-double LSTM model: root mean square error, mean bias error, correlation coefficient, relative root mean square error, Nash–Sutcliffe efficiency, mean square error, mean absolute deviation, coefficient of multiple determination, and Willmott index of agreement. For MSE, RMSE, MAE, MBE, r, R², RRMSE, NSE, and WI, the suggested model improves by 0.000 11, 0.011, 0.008, 0.008, 0.99, 0.98, 2.5, 0.98, and 0.98, respectively. Other machine learning methods, including the Transformer model, single-layer LSTM networks, Autoregressive Moving Average (ARMA) models, Gated Recurrent Unit (GRU) networks, and Bidirectional (Bi-LSTM) networks, are compared to the performance of double LSTM. The twin LSTM model was demonstrated to perform better. The simulations and experimental findings show that the suggested model can precisely estimate wind power. Within the Google Collab environment, the suggested model uses TensorFlow and Keras.

Keywords Wind power · Forecasting · Renewable energy · Deep learning · Optimization

1 Introduction

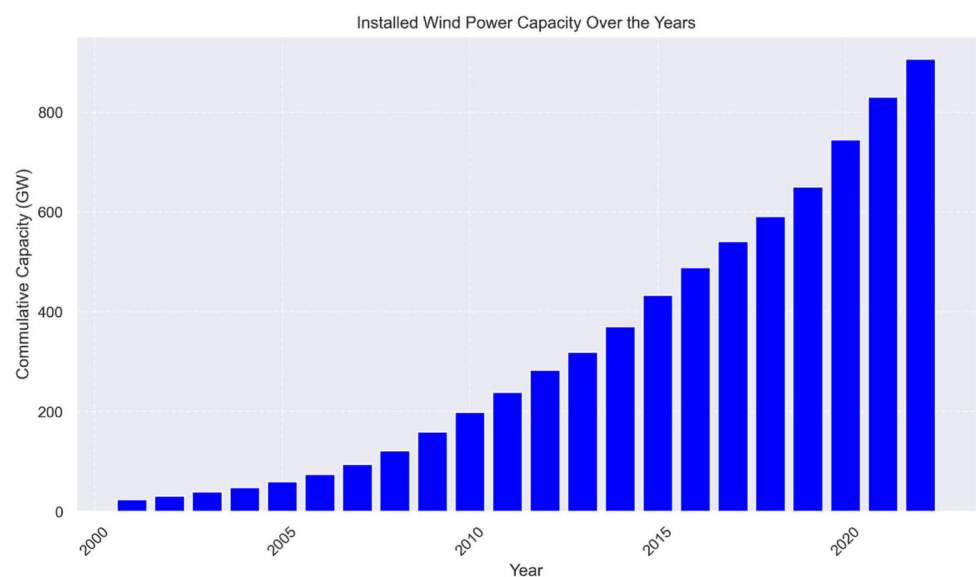
According to data from the International Energy Agency, renewable energy usage standards have increased to 19.3%, while its power generation level is 24.5%. Renewable energy will remain vital to upcoming power consumption patterns as data from the energy sector proves its importance in achieving fossil fuel elimination.



The rise of sustainable power sources such as solar, wind, hydro, and geothermal energy further bolsters this shift, offering a viable alternative to traditional energy systems. Renewable energy systems are suitable fossil fuel alternatives because they function reliably and exist widely throughout the planet. The falling prices to generate and distribute renewable energy enable various countries to adopt it as a sustainable power source that simultaneously protects the environment. The power grid faces obstacles when implementing renewable energy systems mainly because wind energy can be periodic. Predicting wind speeds is vital in maximizing wind energy production and maintaining steady power grid operations. Improvements in wind energy production speed depend directly on flow speed, so better wind speed forecasting algorithms will lead to significant economic and technical developments. Renewable energy development wins, farm layouts, and power transmission requirements heavily depend on applying sophisticated forecasting technologies. Developing methods to forecast wind speed and power generation has consumed considerable research. The forecasting methods are divided into three sections: turbine control in the short-term and monthly and yearly trend prediction in the long term, with dispatching power functions in the mid-term. This research focuses on the third category, proposing a novel forecasting technique for predicting daily mean wind speed over a year. The global wind power capacity has grown substantially, reaching 906 GW as of 2022 (see Fig. 1), reflecting the increasing reliance on renewable energy sources.

Despite these advancements, wind energy forecasting remains challenging due to wind's stochastic nature. Researchers have employed advanced methodologies, including Long Short-Term Memory (LSTM) networks [1], which leverage historical data for precise predictions. Time series forecasting has proven crucial for improving wind energy reliability as it helps decrease fossil fuel use and sustain energy development. The combination of real-time statistical and climate forecasting models increases prediction accuracy according to [2]. However, forecasting remains an ongoing research challenge requiring continuous refinement of methods to improve reliability and stability. This study employs historical time series data analysis to examine wind energy production and utilization [3]. Artificial intelligence (AI), particularly ensemble learning methods, has significantly improved over traditional approaches. Hybrid AI models enhance forecasting accuracy, as demonstrated in Kao et al. [4], where the XGBoost algorithm and a causal sliding window method optimized predictions. However, model interpretability remains critical in machine learning-based forecasting, especially as complex AI models are more challenging to interpret. The problem of AI decision transparency receives its solution through Explainable Artificial Intelligence (XAI) according to [5]. Implementing XAI improves deep learning model performance levels by adding necessary interpretability features that researchers and industry stakeholders need to understand AI-produced predictions. This research integrates LSTM neural networks with XAI tools to improve

Fig. 1 Wind power capacity over a year



energy forecasting efficiency. LSTM models effectively capture temporal dependencies in data but introduce high complexity. By incorporating XAI, we aim to create a transparent framework that clarifies how LSTM processes and generates predictions. Enhancing model interpretability builds user confidence and trust in AI-driven forecasting systems. The main contributions of this work include:

- Utilizing LSTM to explore the search space, preventing suboptimal solutions effectively.
- Employing SHAP, an XAI technique, to explain model performance and improve interpretability.
- Identifying key environmental factors influencing wind energy forecasting results.
- Developing an X-double LSTM model that enhances prediction accuracy across diverse environmental scenarios, mitigating wind power unpredictability and improving grid integration.
- The proposed method requires statistical testing to show how it outperforms current methods.

The rest of this research paper follows this structure: Section 2 reviews existing literature through a summary of previous work. Section 3 describes the framework theory and necessary research methods. Section 4 presents the experimental procedures through which LSTM worked with XAI for energy forecasting assessment. Section 5 shows and interprets the findings through a series of results and analyses. The paper ends by offering a summary of essential discoveries and suggesting new paths for renewable energy forecasting research in Section 6.

2 Related works

This section presents papers that compare wind power prediction models and highlight various methods and developments in the subfield. The studies presented and discussed here in detail explore the various approaches adopted to enhance the efficiency of the conventional techniques employed in wind power prediction. Tian et al [6]. proposed an original WPF system composed of two stages of attention mechanisms, allowing the model to address relevant data points and improve the prediction quality effectively. This establishment in deep learning algorithms forms the basis of the feature and performance inspection. Likewise, New et al. [7] proposed a similar data-driven WPF model that explored high-level deep learning techniques. Their model fared even better than existing models to show a substantial enhancement in performance indicators like Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Mean Squared Error (MSE) to vouch for the capability of their model in precise wind power forecasting. The problem of working with a small amount of historical data in new wind farms was discussed in Dong et al. [8]. Suggested transfer learning in corrupted power curve estimation for wind farms using a small data set was proposed. This method improves the efficiency of few-shot learning by a large margin and can help avoid the problem of a small number of training cases in wind power forecasting. At the same time, He et al [9]. introduced the incorporation of the LSTM models with the Convolutional Neural Networks (CNNs) to enhance short-term WPF performance. When used for a wind farm in northwest China, their work showed LSTM and CNN superior to other sophisticated machine learning algorithms. The scholars Al-Kasabi et al. [10] used variables like wind direction and velocity to improve wind energy forecasting using machine learning. Their model was tested in several sites, particularly France and Turkey, and was found to give reasonable estimates of wind energy production. Shahid et al [11]. used a set of neural networks on genetic programming and an auxiliary predictor. According to a double t-test of their results, the authors pointed out that their model yielded a better prognosis than other methods utilized before. Ban et al [12]. introduced a spatiotemporal graph convolutional network for accurate short-term wind power forecasts. Their model outperformed five conventional forecasting methods based on MAE and RMSE across different forecasting horizons. Residual correction and ensemble reinforcement learning models were developed with the help of Yin and Hui [13], alongside an outlier correction method. Another promising approach, based on Q-learning, is presented as an effective means of forecasting wind power. Peiris et al. [14] used ANN in accurately predicting the wind energy production of the Pawan Danawi wind farm in Sri Lanka at an 85% validation level. On the other hand, Delgado and Fahim [15] employed LSTM networks since, as claimed by the model's authors, LSTM networks

are excellent in capturing nonlinear dynamics to improve the prediction of the performance of wind turbines. López and Arbolea [16], and Sun and Jin [17] also stressed that PCC was widely used for input selection in LSTM and DNNs. Their studies focused on the features that should be utilized to increase the model performance indicators. Xiong et al. [18], advancing the data selection explicitly focused on reconstruction, employed an approach whereby input variables are ranked. A few recent literatures proposed integrated models, where Yu et al. [19] used wavelet decomposition on wind speed data. They divided the data into subseries, which improved the effectiveness of the forecast models, including the Recurrent Neural Networks (RNN), Long Short-Term Memories (LSTM) and Gated Temporal Units (GTU). Subsequently, Shang et al. [20] went a step further and utilized a computational technique known as Complementary Ensemble Empirical Mode Decomposition, which lends enhanced characterization of historical wind speed data and then used a model selection technique known as the Self Organizing Map Clustering for efficient categorization. However, Praveena and Dhanalakshmi [21] went in a different direction, characterizing the days by similar wind conditions with the help of Fuzzy K-means Clustering. They realized that this technique was proper precisely because it allowed them to look for changes in the wind speed over a twenty-four-hour cycle, which gave a handy forecast. Clustering of climatology temperature and humidity parameters of the region was the scope of Ghanbarzadeh et al. [22]. They wanted to design an ANN model for predicting wind speeds. It also had higher levels of prediction accuracy than the other forecasting models employed by the companies. Abdoos et al [23]. proposed an advanced wind power forecasting model using Variational Mode Decomposition (VMD) and Extreme Learning Machine (ELM), integrating the Monte Carlo method for probabilistic forecasts, further enhancing accuracy and reliability. Zhao et al [23]. developed a model combining the LightGBM algorithm and the neural prophet technique, optimizing parameter tuning through grid search, which improved efficiency and reduced computation time. Finally, Mei and Ma's prediction model [24] combined CNN, an attention mechanism, and a bidirectional LSTM network to capture linear and nonlinear relationships in wind power time series. In another research, F.Shahid et al. [25] used a genetic algorithm application for LSTM window size and neuron number optimization to improve the accuracy of wind power prediction compared to existing methods. National Renewable Energy Laboratory's wind power forecasting was enhanced by Azimi et al [26]. integrating the traditional K-means clustering approach with the discrete wavelet transform and the multilayer perceptron neural network. Shi et al [27]. used variational mode decomposition and LSTM to predict hourly wind power for a Chinese wind farm with an accurate day-ahead forecast. A.Lahouar et al [28]. improved a wind power forecasting model based on random forest (RF) and showed that it was more accurate by adding average wind speed and wind direction as the input features. However, adding these features sometimes degrades the results significantly. In another research, Velazquez et al [29]. suggested the impact of wind speed, wind power density, and power output on the performance of ANN models. Using wind direction as an input can reduce forecasting errors. In the present study, Su et al [30]. used wind power prediction analysis to decompose the wind speed data into four low-frequency and four high-frequency. Then, the four high-frequency components were extracted into 60 intrinsic mode functions (IMFs) by ensemble empirical mode decomposition (EEMD). These components were then incorporated into individual LSTM models with yaw error and rotor speed input. The power prediction results of the proposed approach were significantly improved in accuracy. Nevertheless, the role of the direct application of the wind power dataset for prediction still needs to be examined. Zu et al. [31] employed the WPD method to disintegrate wind power times series into three levels. The gained subseries were given to a gated recurrent unit (GRU), and the predictions were reconstructed to get the results. Experimental analysis revealed that the proposed WPD-GRU-SELU model is more accurate for prediction than existing RNN models. In another research, Mujeeb et al. [32] used Wavelet Packet Transform (WPT) with Deep convolutional neural network (DCNN) to forecast the day-ahead hourly wind power of ISO New England's wind farm; nevertheless, the authors failed to forecast the subseries using different independent models. Besides wind power prediction, other wavelet transform methods have recently been employed in the wind power prediction domain. However, they identified the need for improved model interpretability. Therefore, this paper has provided an exhaustive chronology of the wind power forecasting techniques developed in chronological succession, advanced neural networks, hybrid wind power forecasting models, and

methods of the scarcity of data and predictive accuracy. Again, these works indicate that work is ongoing to enhance the precision of wind power forecasts since such enhancements are critical in handling wind power systems. Last, Table 1 revisits the considered investigations and identifies their impact on developing forecasting methodologies for wind power systems.

3 Materials and methods

These are the methodologies employed in the preprocessing and modeling stage from the research highlighted in this study. The data was cleaned adequately before analyzing the collected data to ensure datasets were correctly compiled for analysis. Regression analysis procedures, such as imputation and removal of outliers, were applied to improve the data quality. The main objective focused on maintaining a comprehensive feature collection alongside optimizing model operational speed. Our team used an amin-max scaler because this method provides specific value ranges from 0 to 1 for data interpretation and comparison in wind power generation forecasting tasks. After getting the data in shape, we laid out the outcomes with various prediction plots for selected cases.

Table 1 Summary of Wind Power Forecasting (WPF) Studies

Authors	Technique	Results	Limitations
Tian et al. [6]	Two-stage attention mechanism in deep learning	Superior accuracy and performance	Limited validation across different locations
New et al. [7]	Data-driven deep learning model	Reduced MAE, MAPE, and MSE	Performance may degrade with imbalanced data
Dong et al. [8]	Transfer learning for data scarcity	Improved few-shot learning accuracy	Highly dependent on available small datasets
He et al. [9]	LSTM and CNN combination for short-term WPF	Outperformed traditional models	Sensitive to noisy data
Al-Kasabi et al. [10]	Ensemble models with wind direction and speed	Effective in France and Turkey	Requires extensive meteorological data
Shahid et al. [11]	Neural networks with genetic programming	Better than existing techniques	Limited generalization due to dataset specificity
Ban et al. [12]	Spatiotemporal graph convolutional network	Improved MAE and RMSE	High dependency on spatial data availability
Yin & Hui [13]	Residual correction and reinforcement learning	Significant improvement in Q-learning-based forecasting	High computational cost
Peiris et al. [14]	ANN for wind energy prediction	85% accuracy in Sri Lanka	Results may not generalize well
Delgado & Fahim [15]	LSTM for turbine performance	Captured nonlinear dependencies	Requires large datasets
López & Arboleya [16]	PCC analysis for LSTM and DNN	Identified relevant input variables	PCC may not always be optimal
Yu et al. [19]	Wavelet method for wind data	Improved hybrid model performance	Sensitive to sudden changes
Shang et al. [20]	Ensemble empirical mode decomposition	Categorized data efficiently	High computational cost
Praveena & Dhanalakshmi [21]	Fuzzy K-means clustering	Improved daily pattern forecasting	Struggles with dynamic wind patterns
Ghanbarzadeh et al. [22]	ANN using meteorological variables	High wind speed prediction accuracy	Sensitive to climate variations
Abdoos et al. [23]	VMD and ELM with Monte Carlo	Accurate probabilistic forecasts	Computationally intensive
Zhao et al. [23]	LightGBM and Neural-Prophet combination	Improved interpretability	Parameter tuning is time-consuming
Mei & Ma [24]	CNN, attention mechanism, and bidirectional LSTM	Captured linear and nonlinear relationships	Complexity time hinders real-use

These visualizations demonstrate how well our modeling approach forecasts wind power generation, so they are not merely for show. We will also give you a thorough assessment of the study performance, which will help you understand how accurate and dependable our model is. Our fundamental objective is simple: we use several sophisticated deep learning techniques, such as the Transformer model, single-layer LSTM networks, and double layer

LSTM networks, Autoregressive Moving Average (ARMA) models, Gated Recurrent Unit (GRU) networks, and Bidirectional (Bi-LSTM) networks, to forecast wind power generation. We're not limiting ourselves. We're rigorously testing the performance of a diverse range of models, including the Transformer model, single-layer LSTM networks, double layer LSTM networks, Autoregressive Moving Average (ARMA) models, Gated Recurrent Unit (GRU) networks, and Bidirectional (Bi-LSTM) networks. This paper evaluates these analytical methods to determine their predictive potential and ability to represent wind power generation intricacies. The team pursues quantitative analysis and qualitative evaluation to study our research process thoroughly. Our analysis begins with investigations of the dataset to understand the patterns alongside visualization studies. Data collection proceeds through partitioning into training data and testing data to build reliable models during development. Five distinct models go through training as part of this process to provide independent deep learning methods. We apply the reserved data set to evaluate our trained models during testing. A deep learning model's performance evaluation determines the optimal solution. Figure 2 displays a complete diagram of the proposed wind forecasting technique that covers the entire process, starting with data preparation and ending with model assessment.

3.1 Dataset

We got our hands on a publicly available wind dataset from Kaggle, packed with essential information for studying and analyzing the factors and features. This data was pulled from four major German energy companies: 50Hertz, Amprion, TenneT TSO, and TransnetBW. It shows non-normalized power generation figures recorded every 15 minutes, giving us 96 daily data points. The dataset covers the period from August 23, 2019, to September 22, 2020, and offers a thorough foundation for examining swings in wind energy output over a little more than a year. Figure 3 shows the complex interrelationships among these critical factors and how each affects

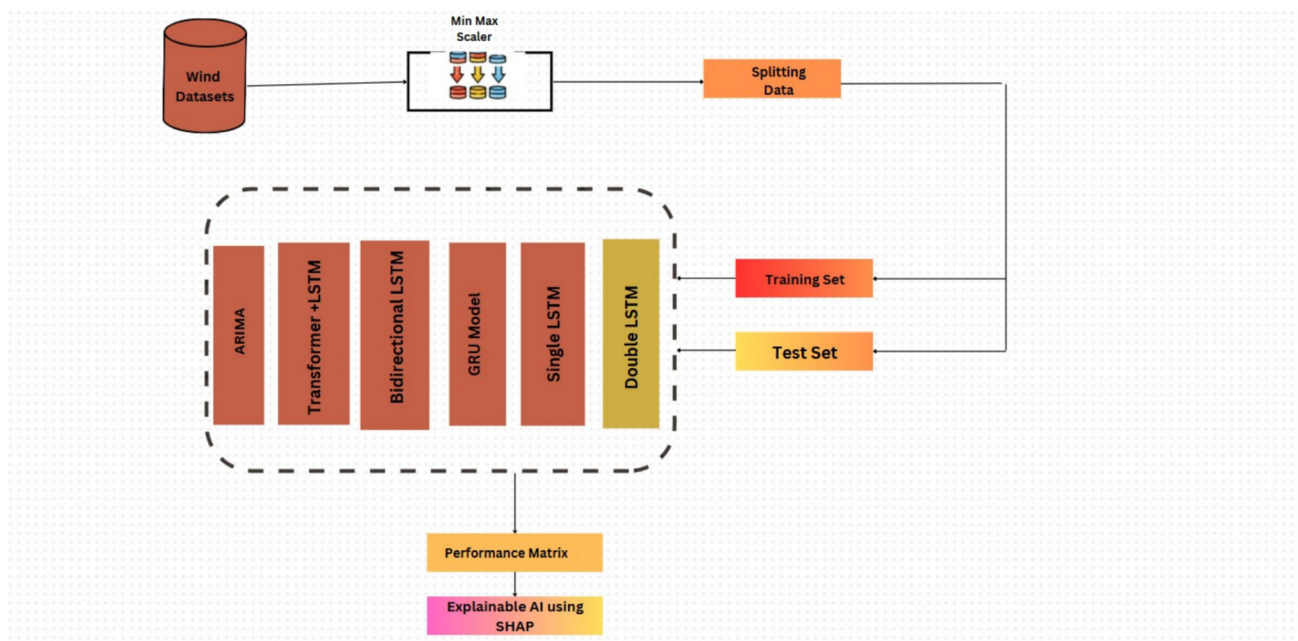


Fig. 2 Block schematic of the suggested methodology

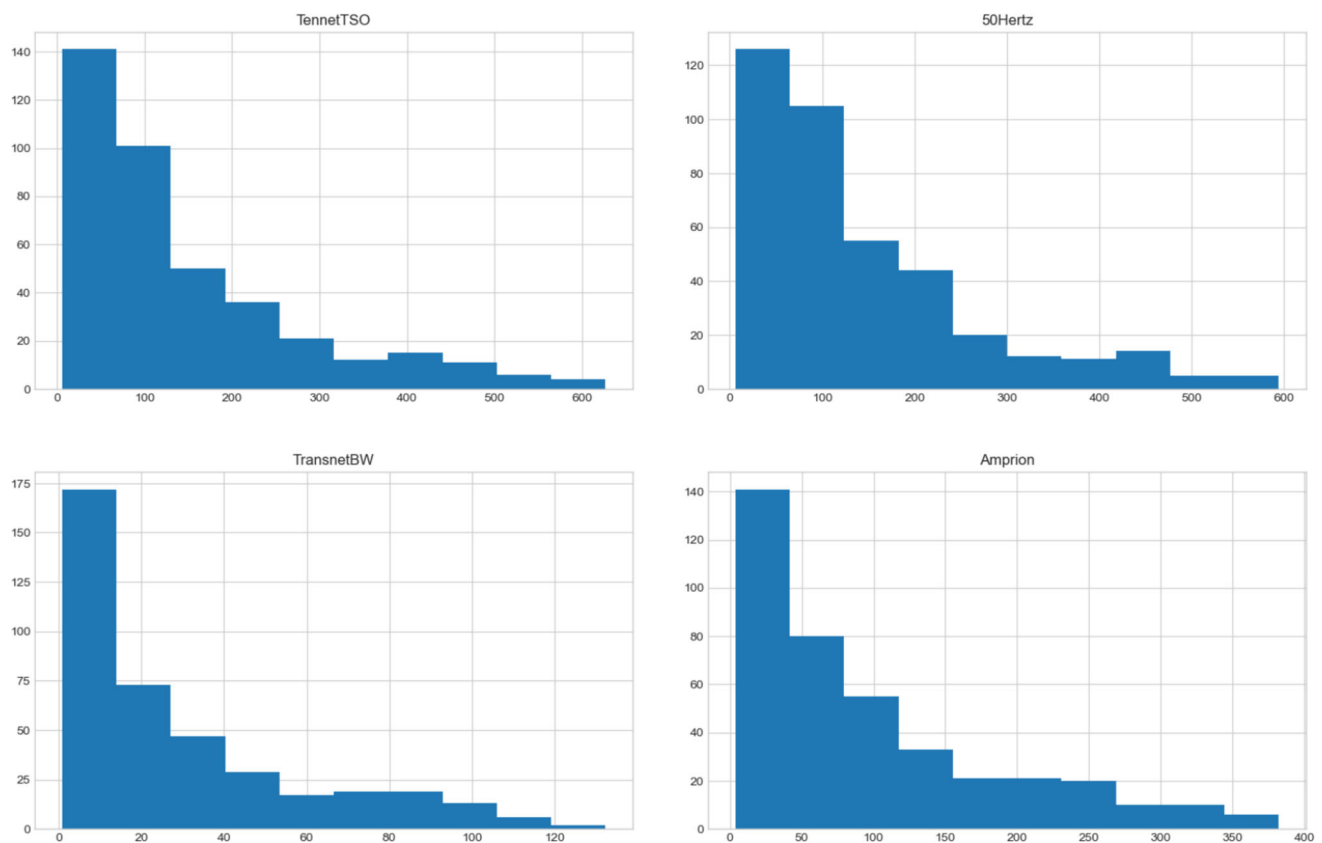


Fig. 3 Graphical representation of the histogram for every feature in the energy efficiency dataset

the total amount of wind energy produced. This visualization can help guide future energy policies and improve comprehension of dynamics involved in wind energy production.

Research variables produce distinct relationships and patterns from the data, which helps us understand their interdependent behavior. High numbers of interdependencies within wind energy forecasting prove why developing an accurate prediction model is essential. A credible model proving essential for forecasting becomes essential because of the pressing need to improve our energy management approaches and policy and strategic decision-making. Developing such a forecasting model will not only enhance our understanding of the given data set and its forecast but also advance our analysis of the patterns of renewable energy production. From our detailed study, these insights are unique and crucial for any stakeholders in wind energy to engage in sound planning and action to maximize energy yield and incorporate wind energy into the overall power network. Figure 8 showcases the elaborate relationships between the four German transmission system operators TenneT TSO, 50Hertz, TransnetBW, and Amprion using heat mappings. Visual representation of operator correlation values between 0 and 1 makes the heatmap reveal significant data relations that keep operators relevant. Lighter shades of red indicate weaker associations, while darker shades represent stronger correlations. For instance, TenneT TSO and 50Hertz show a high degree of similarity with a correlation of 0.87, while 50Hertz and TransnetBW have a moderate connection with a correlation of 0.61. Overall, the heatmap reveals significant relationships between the operators that could be harnessed for collaborative efforts to enhance energy management and grid stability (Figs. 4 and 5).

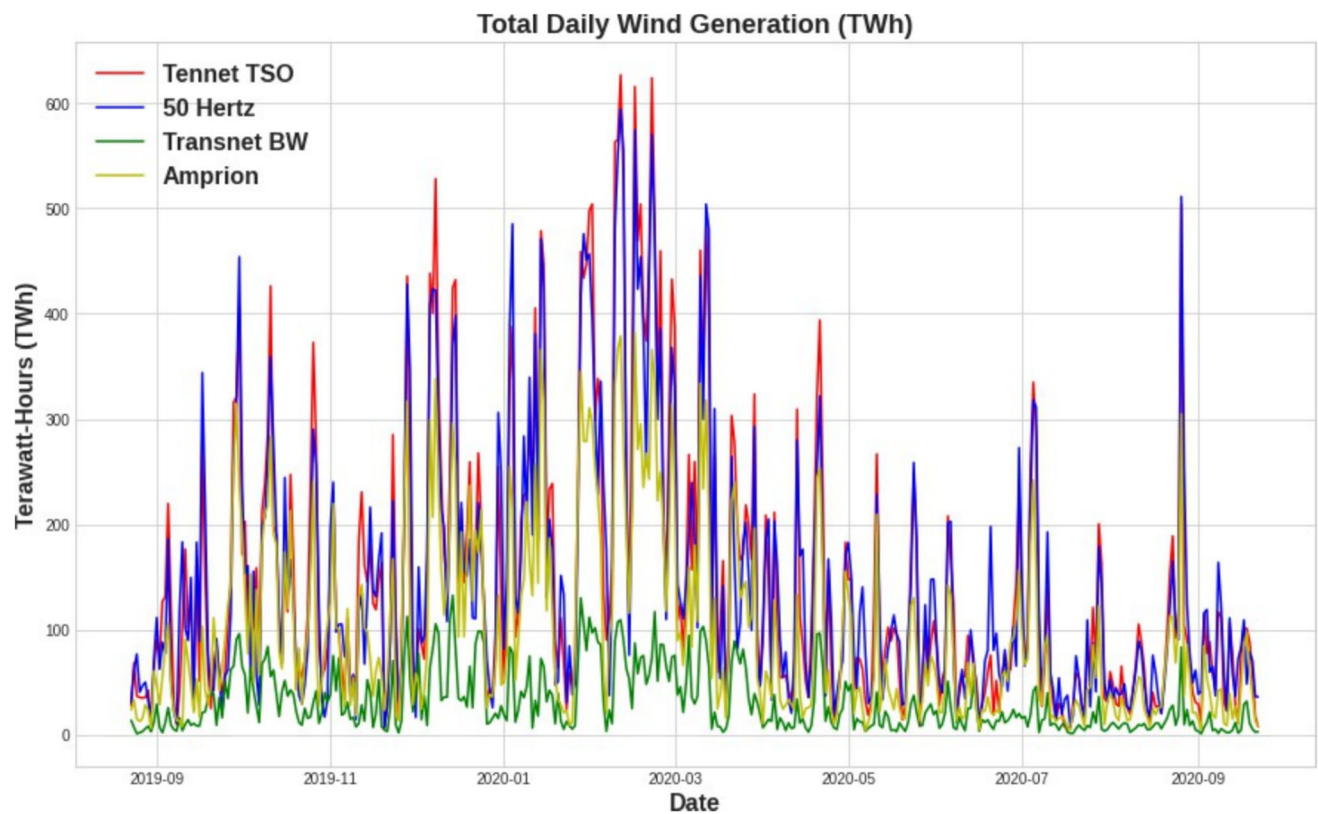


Fig. 4 Total daily wind power generation over time

Fig. 5 Heatmap graph for dataset attributes related to wind energy efficiency



4 Deep learning (DL) models

4.1 Long short-term memory (LSTM)

To effectively manage and reduce the significant correlations that often afflict nonlinear regression tasks, LSTM models are sophisticated neural network designs. Sequential data regression applications, like time series analysis, are incredibly remarkable because of this. Unlike traditional statistical methods that mainly focus on predicting outcomes based on the immediate past of a stochastic process, LSTM models have the remarkable ability to

remember and integrate information about events that occurred way back, even long before the query was made. This distinctive feature enhances their ability to estimate the probability of future occurrences, making LSTM a great powerhouse tool for predicting scenarios. Based on the work [33], this ability is efficient for time series regression forecasts as it allows the model to influence predictions by previously observed time points, significantly improving the quality of the results. One study reveals that original classical LSTM methods perform better than other regression task approaches. This is particularly the case in financial modeling, traffic prognostication and meteorology. This paper underscores the effectiveness of LSTMs, especially in regression problems with the time series [34] data and specific implications of optimum strategies and design changes for its implementation. In addition, what makes LSTM unique is the hidden layer in its structure. The LSTM network consists of a memory cell (cell state), forget gate, input gate, and output gate [35]. The formula [36] for the LSTM component is as follows:

$$a_t = \sigma(W_a[q_{t-1}, z_t] + b_a) \quad (1)$$

$$s_t = \sigma(W_s[q_{t-1}, z_t] + b_s) \quad (2)$$

$$\hat{C}t = \text{ReLU}(W_c[q_{t-1}, z_t] + b_c) \quad (3)$$

$$C_t = a_t \cdot C_{t-1} + i_t \cdot \hat{C}t \quad (4)$$

$$o_t = \sigma(W_o[q_{t-1}, z_t] + b_o) \quad (5)$$

$$q_t = o_t \cdot \text{ReLU}(C_t) \quad (6)$$

$$\text{ReLU} : f(z) = \max(0, z), \quad (7)$$

where a_t is the forget gate, s_t is the input gate, Ct is the newly added value to the cell gate, C_t is the cell state, o_t is the output gate, and q_t is the output value at time t . The activation functions used are σ (sigmoid) and ReLU. Here, W_s , W_c , and W_o represent the weight matrices for the respective gates. q_{t-1} is the output value at time $t - 1$, C_{t-1} is the cell state at time $t - 1$, x_t is the input value at time t , and b_a , b_s , b_c , b_o are the bias values for the respective gates.

4.2 Bidirectional (Bi-LSTM)

Recent advancements in RNNs' have enhanced the development of models capable of processing forward and backward data. Predictability of more than 90 percent was obtained by tri-training the authors' models of bi-LSTMs and bidirectional Gated Recurrent Units (Bi-GRUs), respectively; these architectures are particularly suitable for time series forecasting [37, 38]. They improve their capacity to sort data into categories and foretell based on past and future data. Bidirectional Long and Short-Term Memory Networks (bi-LSTMs) are also bidirectional gated recurring units (bi-GRUs) used in natural language processing since their inception in 1997. Due to their flexibility, they can efficiently perform tasks as simple as classification or sorting of text. Here, sentiment analysis is all about analyzing the sentiment projected to a text, while language translation is about translating content from one language to another. The application efficiency of such models increases their significance and performance in managing real-life sequential data [39]. In bidirectional LSTM, the working of the forward layer is the same as that of the standard LSTM, which traverses from time $t = 1$ to T . However, in the case of the backward layer, the input sequence is fed from $t = T$ to 1. Therefore, a mathematical representation of

the LSTM cell in the forward layer will remain the same. In contrast, mathematical expressions of the LSTM cell in the backward layer denoted by the leftward arrow ($\leftarrow$) at time t can be written as [40]:

$$\vec{F} = \sigma(W_f \vec{X}_t + W_f \vec{R}_t + i), \quad (8)$$

$$\vec{I} = \sigma(W_i \vec{X}_t + W_i \vec{R}_t + i), \quad (9)$$

$$\vec{O} = \sigma(W_o \vec{X}_t + W_o \vec{R}_t + i), \quad (10)$$

$$\vec{G} = \tanh(W_g \vec{X}_t + W_g \vec{R}_t + i), \quad (11)$$

$$\vec{C}_t = \vec{F} \odot \vec{C}_t - 1 + \vec{I} \odot \vec{G}, \quad (12)$$

$$\vec{R}_t = \vec{O} \odot \tanh(\vec{C}_t). \quad (13)$$

At each time step of bidirectional LSTM, two outputs will be generated, one by forwarding the LSTM layer (R_t) and one by the backward LSTM layer ($R_t \rightarrow$). These two outputs will be merged to get the output. Final output Y will be expressed as combining the outputs generated by individual LSTM layers.

$$Y_t = R_t + \vec{R}_t, \quad (14)$$

$$Y = y_{t-1} + y_t + y_{t+1}. \quad (15)$$

4.3 Gated recurrent unit (GRU)

The Gated Recurrent Unit, like the LSTM, is a Recurrent Neural Network. It, however, has a less complex architecture than LSTM-RNN. Further, in 2014 [41], proposed an improved architecture of RNNs in Gated Recurrent Units (GRUs), which were further enhanced over the standard LSTM networks. They can handle streams of identical input sequences of any length, the same as LSTMs, and hold the state containing historical info. Yet, to control the flow of information, there is only a single update gate to determine which data to let through and which to reject and a reset gate to decide when to forget the past and start anew. This is not the same as the LSTMs, which work with several gates and an internal memory cell. For this reason, GRUs are less complex than LSTMs and can be trained much more straightforwardly but are equally effective when used in different tasks [37]. As postulated by [37], GRUs have better performance compared to LSTMs in language modeling than that of Penn Treebank. Indeed, in an NLP model comparison, GRUs again performed better than CNNs and LSTMs in several qualities, ensuring the domination of GRUs. However, GRUs can outperform ordinary RNNs in identifying the long dependency in sequential data patterns. Even though GRUs control current data differently based on the current input and network state, their update and reset gates selectively enable the reminiscence or forgetfulness of earlier information. Among the types of RNNs used for tasks, where it is necessary to store and use a lot of information within the sequences, the so-called GRU is the best choice for you. GRU has fewer gates than LSTM, which contributes to its high efficiency. The transition functions of GRU are formulated as follows:

$$r_t = \sigma(x_t W_r + h_{t-1} W_r + b_r), \quad (16)$$

$$z_t = \sigma(x_t W_z + h_{t-1} W_z + b_z), \quad (17)$$

$$\tilde{h}t = \tanh(x_t W_x + (r_t \odot h_{t-1}) W_h + b), \quad (18)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}t. \quad (19)$$

4.4 Autoregressive integrated moving average (ARIMA)

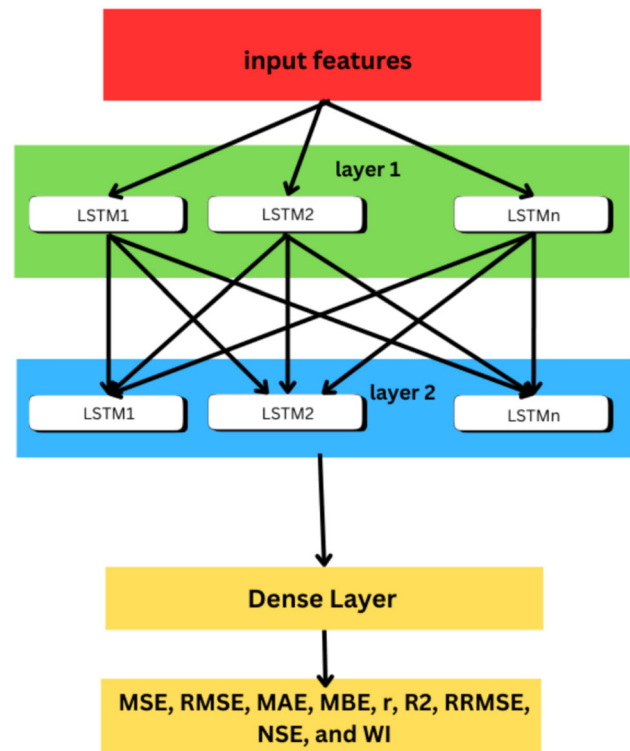
ARIMA, which stands for AutoRegressive Integrated Moving Average [42], is a robust statistical model commonly used for time series analysis. This model is designed to analyze data collected at regular intervals, allowing for a consistent approach to forecasting. By leveraging historical data, ARIMA aims to predict future values based on observed patterns and trends. The model focuses on two primary variables: the predicted characteristics, such as sales, temperature, or stock prices, and the time these predictions are made. ARIMA consists of two essential components: the autoregressive (AR) part, which uses the relationship between an observation and several lagged observations (past values), and the moving average (MA) part, which models the interaction between an observation and a residual error from a moving average model. To effectively implement an ARIMA model, three critical hyperparameters must be identified and fine-tuned: The number of autoregressive lags (p): This parameter identifies the amount of previous stale observations regarding integration into the model. Choosing a proper p value lets us describe the current value's dependency on its historical counterparts. The order of differencing (d): This parameter refers to the number of times the data has to be differenced to become stationary. As it can be recalled, one of the primary assumptions or conditions in using the ARIMA models is stationarity to guarantee that the properties of the series are stable over space. The number of moving average terms (q): Out-of-home moving expenses. This parameter determines the number of lagged forecast error terms in the forecast equation. This is much easier said than done, and q must be appropriately determined to accommodate the noise in the data. If well chosen, these hyperparameters allow practitioners to obtain an improved ARIMA model for better forecasts based on historical time series data. In this context, the next step remains to identify the ARIMA model (p, d, q) to fit the situation under consideration. The long-standing time meaning that the esteem of a variable could be a straight mixture of past values and past errors, which is shown in the ARIMA model as follows:

$$X_t = \varphi_1 X_{t-1} + \varphi_2 X_{t-2} + \cdots + \varphi_p X_{t-p} + m_t - \theta_1 m_{t-1} - \theta_2 m_{t-2} - \cdots - \theta_q m_{t-q},$$

where X_t is the actual value, m_t is a random error at time t , $\varphi_1, \varphi_2, \dots, \varphi_p$ are the coefficients of the autoregressive (AR) model, and $\theta_1, \theta_2, \dots, \theta_q$ are the coefficients of the moving average (MA) model. The parameters p and q are integers representing the orders of the autoregressive and moving average components, respectively.

4.5 Proposed model based on XAI

During training, a parametrized machine learning algorithm processes artificial data to optimize the minimization of objective functions through adjustable parameters. This phase constitutes the LSTM's data supply method through batch distribution. The LSTM parameters receive their definitions and then proceed with evaluation in the next operation phase. The illustration in Fig. 6 presents the LSTM Architecture. By applying the SHAP tool, users can obtain an understandable model that is interpretable to neighborhoods after completing the training process. Current AI systems generate user concerns through black-box models by showing prediction outputs,

Fig. 6 LSTM architecture

while remaining completely opaque about variable-input relationships to output results. Interpretable models, including linear regression, demonstrate superiority because they disclose the strength of variables to influence prediction results. XAI provides a solution that enables people to understand the assessment processes behind AI systems. The search for how features and model components affect each other occurs through black-box methods in XAI during training periods. Domain experts in photovoltaic power generation benefit from the solution because it generates results that allow users to evaluate model rationality and follow variable interactions. SHAP defines itself as the leading approach for delivering predictions from machine learning models through its adaptable methodology. The explanation technique works with any model type because it lacks specific model requirements, thus allowing its application across tree-based models, neu, real networks, and linear models. SHAP derives its feature contribution calculations through the implementation of Shapley values. The distribution of benefits occurs through the Shapley value, which emerged from cooperative game theory to give equal shares to coalition players. Researchers have observed significant expansion in adopting this theory to create machine learning explanation models throughout the past few years. The model explanation system displays both variables as player characteristics through features and predictions as revenue outputs, while adopting principles of Shapley value theory. The mathematical calculation for the Shapley value requires the following formula.

$$M_j = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!(n - |S| - 1)!}{n!} \times (x(S \cup \{j\}) - x(S)). \quad (20)$$

where M_j is the contribution of the j -th variable; N is the set of all players (features) $(1, 2, 3, \dots, j, \dots, n)$, which is the complete set; S is a subset of N , in which the explained feature j is removed, with a total of 2^n ; v is the gain function:

$$x(S) = EA\hat{f}(y)|_{y_S}, \quad (21)$$

where $\hat{A}$ is the empirical distribution of the training data; and f is the black-box model. SHAP assesses feature significance through global and local explanations, demonstrating both the general effects on model prediction and the individual instance outcome impacts of each variable. SHAP has rapidly become popular within different application domains, including LT care, image processing, and other areas. System transparency increases due to understandable explanations regarding advanced machine learning models through the SHAP method, allowing users greater confidence in their predictions.

5 Experimental results

The evaluation of wind power generation prediction models was conducted using several key metrics: Mean Square Error (MSE), Mean Absolute Deviation (MAD), Coefficient of Multiple Determination (R^2), Root Mean Square Error (RMSE), Mean Bias Error (MBE), Correlation Coefficient (r), Relative Root Mean Square Error (RRMSE), Nash-Sutcliffe Efficiency (NSE), and Willmott Index of Agreement (WI). Table 2 presents the metrics that evaluate the prediction models according to their efficiency and accuracy. The computational duration for each model training was considered since it determines the practical implementation potential of this proposed method. Since each model involves a unique computational cost and exhibits varying strengths and weaknesses in terms of accuracy, the selection process is based on a trade-off between these characteristics. Tables 3 and 4 provide a comparative analysis of various machine learning models to highlight their advantages and limitations in different conditions for wind power forecasting. Among these models, the Double Layer LSTM is the best performer across all evaluation metrics. As shown in Tables 3 and 4, it achieves the lowest MSE (0.000113) and RMSE (0.011021), indicating superior accuracy with minimal prediction errors. Furthermore, its correlation coefficient ($r = 0.99996$) supports an almost perfect prediction capability. The model displays robust properties through NSE (0.98762) and WI (0.98465) values indicating reliability. The Double Layer LSTM reaches an optimal rate of accuracy and efficacy through its 9.35-second computation time. Competitively, the Single Layer LSTM and GRU models achieve performance metrics slightly worse than the Double Layer LSTM results. The single-layer LSTM achieves an MSE of 0.000199 along with an RMSE of 0.018743, and the GRU model matches these results with an MSE of 0.000198 and an RMSE of 0.019056. Predictive accuracy remains satisfactory for these models since they demonstrate correlation values that fall between 0.98 and 0.99. The implementation time of the GRU exceeds the single-layer LSTM period, offering 14.39 seconds as opposed to 11.37 seconds. The bidirectional LSTM model demonstrates inferior performance in terms of prediction accuracy

Table 2 Equations for evaluation metrics

Metric	Equation
Coefficient of Determination	$R^2 = 1 - \frac{\sum_{j=1}^n (F_j - Z_j)^2}{\sum_{j=1}^n F_j - F^2}$
Mean Square Error	$MSE = \frac{1}{n} \sum_{j=1}^n (F_j - Z_j)^2$
Mean Absolute Error	$MAE = \frac{1}{n} \sum_{j=1}^n F_j - Z_j$
Root Mean Square Error	$RMSE = \frac{1}{n} \sum_{j=1}^n (F_j - Z_j)^2$
Correlation Coefficient	$r = \frac{q \sum_{i=1}^s (f_i - \bar{f})(\hat{f}_i - \bar{\hat{f}})}{\sum_{i=1}^s (\hat{f}_i - \bar{\hat{f}})^2 \sum_{i=1}^s (f_i - \bar{f})^2}$
Relative Root Mean Square Error	$RRMSE = \frac{\frac{1}{s} \sum_{i=1}^s (f_i - \hat{f}_i)^2}{\bar{f}}$
Nash–Sutcliffe Efficiency	$NSE = 1 - \frac{\sum_{i=1}^s (f_i - \hat{f}_i)^2}{\sum_{i=1}^s (f_i - \bar{f})^2}$
Willmott Index of Agreement	$WI = 1 - \frac{\sum_{i=1}^s (f_i - \hat{f}_i)^2}{\sum_{i=1}^s \hat{f}_i^2 + f_i - \bar{f}^2}$

Table 3 Performance Metrics Depend on MSE, RMSE, MAE, MBE, and r

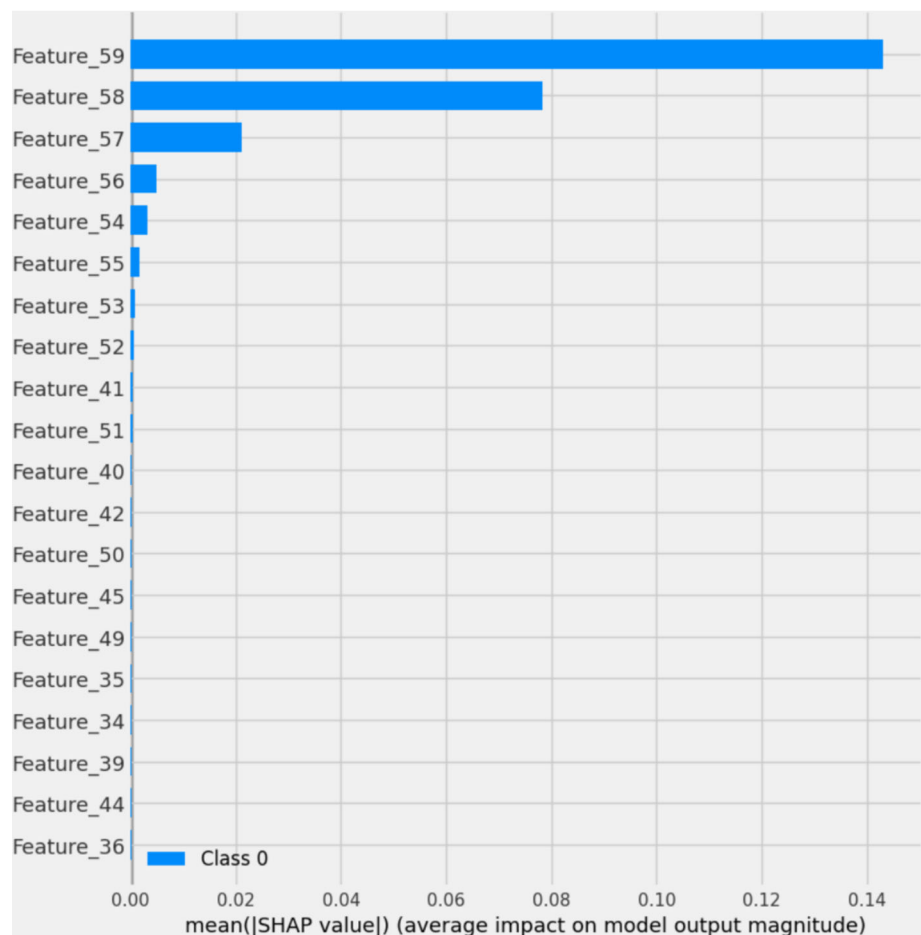
Models	MSE	RMSE	MAE	MBE	r
Double Layer LSTM	0.00011	0.01102	0.00816	0.00843	0.99996
Single Layer LSTM	0.00020	0.01874	0.00962	0.01042	0.98998
GRU	0.00020	0.01906	0.01091	0.01068	0.97993
Bidirectional LSTM	0.00016	0.02174	0.00962	0.03392	0.96986
Transformer + LSTM	0.01959	0.13998	0.09245	0.02912	0.93119
ARIMA	0.02451	0.15655	0.13731	0.01646	0.92137

Table 4 Performance Metrics Depend on RRMSE, NSE, WI, and Fitted Time

Models	R^2	RRMSE	NSE	WI	Fitted Time (s)
Double Layer LSTM	0.98860	2.57637	0.98762	0.98465	9.35368
Single Layer LSTM	0.98730	2.63750	0.97759	0.98276	11.37579
GRU	0.98460	2.70348	0.97756	0.98300	14.39658
Bidirectional LSTM	0.98730	2.80976	0.97641	0.98155	16.41367
Transformer + LSTM	0.93460	2.76483	0.96756	0.98300	17.42966
ARIMA	0.92480	2.89890	0.95479	0.98155	24.46384

compared to the basic LSTM models. The prediction performance becomes less effective because the MSE reaches 0.000162 and the RMSE reaches 0.021743 for this model. Although the MBE value is 0.033915, this model may display an existing prediction bias. The Transformer + LSTM hybrid model demonstrates limited effectiveness for wind power generation prediction according to the data in Table 4. The hybrid model with Transformer + LSTM reaches 0.019594 MSE and 0.139978 RMSE but fails to match LSTM-only models. The combination of Transformer with LSTM does not seem to function well for this particular task. This model needs too much time, at 17.42 seconds, to function practically. ARIMA produces the least successful results among all prediction models. It reports the highest error values, with an MSE of 0.024507, an RMSE of 0.156546, and a lower correlation coefficient ($r = 0.9214$), indicating a weaker relationship between predicted and actual values than deep learning models. Although ARIMA exhibits relatively low MBE (0.04197) and moderate NSE (0.95479) and WI (0.98155) values, its ability to capture wind power generation dynamics remains limited due to its lack of adaptability to temporal patterns. Furthermore, its computation time of 24.46 seconds makes it the least efficient model in this study. Although the bidirectional LSTM model remains a standard tool in many applications, it displays room for enhancement by reducing its current error standards compared to basic LSTM models. The results demonstrate an MSE of 0.000162 and an RMSE of 0.021743, while displaying an MBE value of 0.033915, which indicates possible bias in prediction. Although incorporating two robust architectures, the Transformer + LSTM hybrid model performs less effectively than LSTM-only structures. It achieves an MSE of 0.019594 and an RMSE of 0.139978, but these values indicate a drop in predictive performance compared to simpler LSTM variants. This suggests that the Transformer component may not integrate efficiently with LSTM for this specific task. Furthermore, with a computation time of 17.42 seconds, this model is less efficient than its LSTM-only counterparts. The ARIMA model demonstrates the weakest predictive capabilities at the lower end of the performance spectrum. It reports the highest error values, with an MSE of 0.024507 and RMSE of 0.156546. The correlation coefficient value ($r = 0.9214$) reveals that the predictive strength between actual and predicted values is lower than deep learning models, thus indicating weaker predictive power. Although ARIMA exhibits relatively low MBE (0.04197) and moderate NSE (0.95479) and WI (0.98155) values, its ability to capture wind power generation dynamics remains limited due to its constraints in modeling temporal dependencies. Moreover, its computational time of 24.46 seconds makes it the least efficient model in this study. SHAP (Shapley Additive exPlanations) values were employed further to analyze input features' influence on model predictions, as shown in Fig. 7. Each input feature is represented as a vertical bar, with its position indicating its Contribution to the model's predictions. The SHAP summary bar chart evaluates features through their mean absolute SHAP values

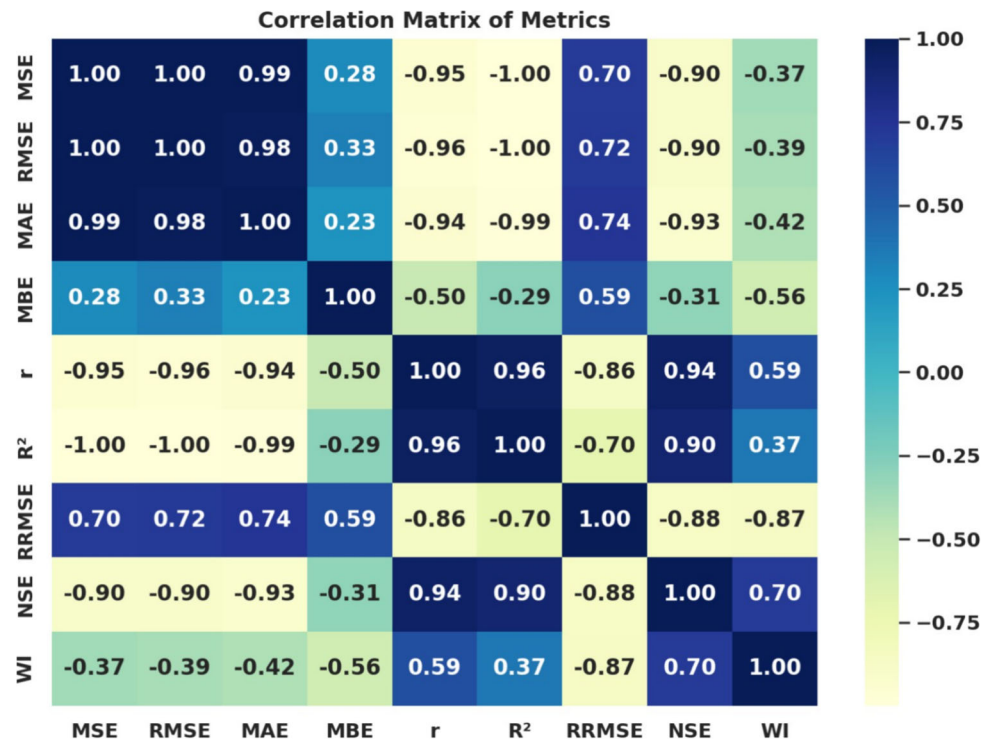
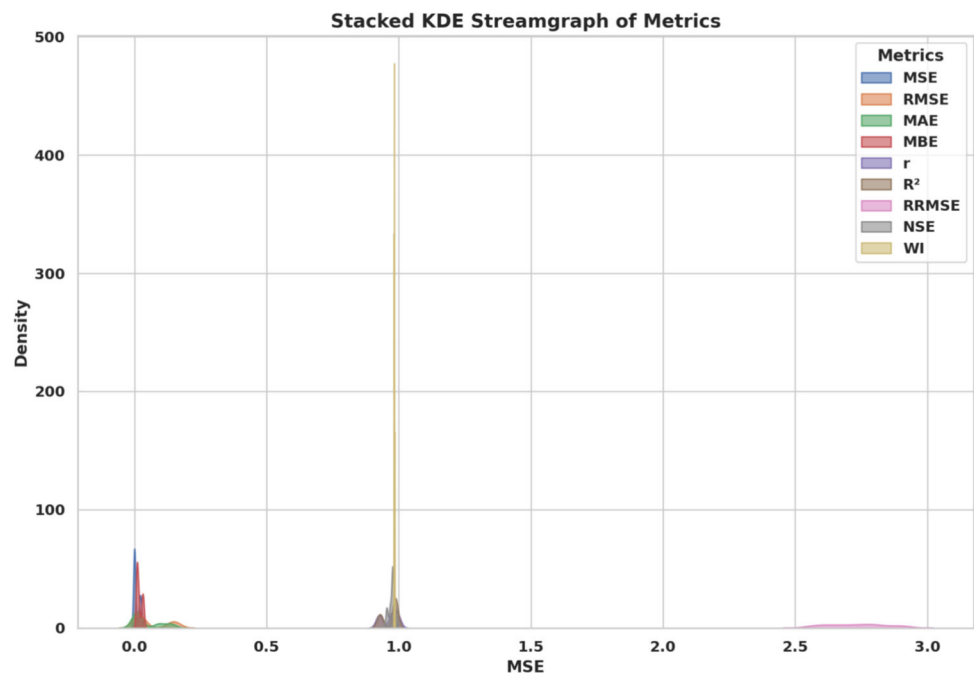
Fig. 7 Contribution of each model attribute to the prediction outcomes



to present the most influential variables for forecasting purposes. The feature values align with the horizontal axis, yet their priority levels are shown in the vertical bars. Through this analysis, researchers gain insights regarding their present model execution and develop a framework for future model optimization to construct effective and shallow models that retain accuracy levels. The data in Fig. 7 reveals that Feature_59 establishes itself as the most influential variable with a score value of 0.76. The significance score of Feature_58 comes in second to Feature_59, according to the summary results. The significance levels start from Feature_57 at 0.88 before reaching a moderate value for Feature_56, then continue declining for less significant elements beginning with Feature_54. This framework produces important results about which features have the most significant impact on predictive values, enabling a better selection of features and reduction methods. Excessive optimization of core variables enables dimensional search space reduction, which might lead to effective shallow models with high accuracy levels.

6 Discussion of statistical result analysis

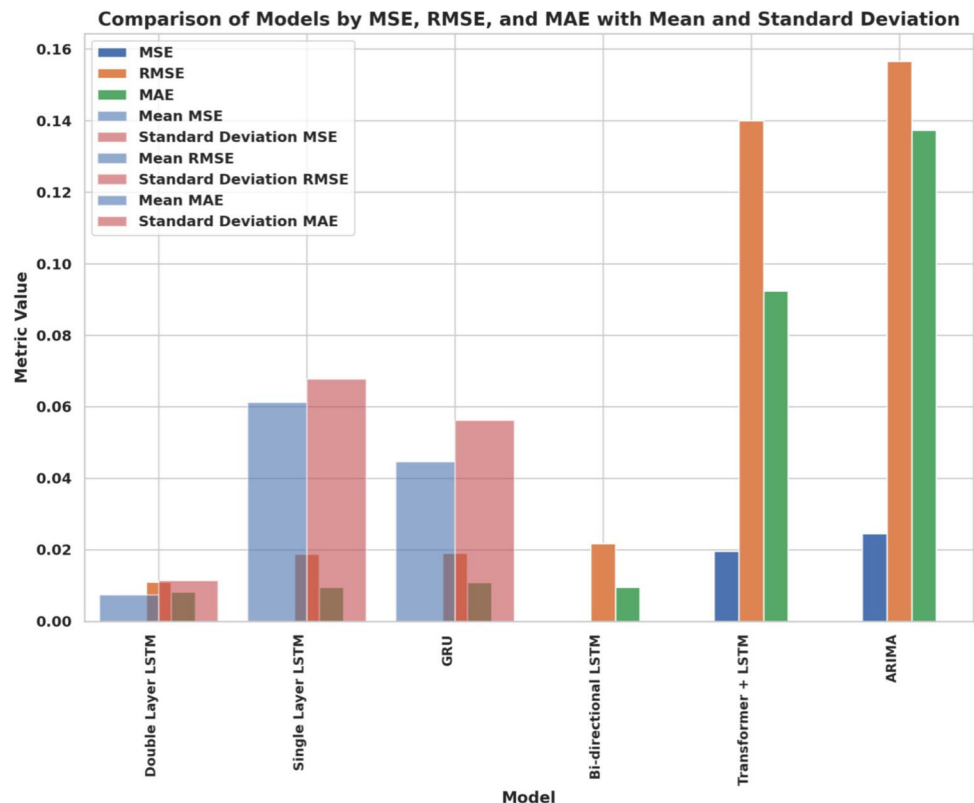
The performance metrics in Fig. 8 are compared using a visual heatmap structure, which includes MSE, RMSE, MAE, MBE, R-squared, NSE, and WI. The performance metrics in the heatmap reveal better results through dark colors, thus enabling easy and rapid detection of the leading models. The proposed Double Layer LSTM demonstrates exceptional performance, particularly in its superior predictive accuracy in almost every evaluation metric. The visual representation exhibits the comparison between models and ultimately demonstrates the Double- Layer LSTM's impressive predictive accuracy, convincing the audience of its effectiveness. Figure 9

Fig. 8 Heatmap of performance metrics**Fig. 9** Stacked KDE of different models

presents a vertical bar chart showing different performance indicators dispersed across models to provide significant statistical data about their statistical matching with the available dataset. Error measures, including MSE, RMSE and MAE, exhibit most of their values distributed in the lower parts of the scale. The model performance shows good results through this concentration of data points that support its reliability through low error margins. Multiple visualizations produce an extensive and comprehensive performance evaluation of the Double Layer LSTM model, providing a wealth of knowledge in wind power forecasting scenarios. The Willmott Index (WI) operates crucially for forecasting accuracy because its value stays at one, demonstrating high agreement between

predicted and actual values. The index functions as a dependable method for evaluating the prediction capabilities of developed models. Figure 9 contains vital points that reveal how different models perform based on their metrics. The visual representation shows that measurement performance between models is comparable because metrics values are distributed symmetrically at low density for errors and high density for WI metrics. This Fig. 10 compares the performance of different models based on three main metrics. Their mean and standard deviation have also been calculated and presented: MSE, RMSE, and MAE. Specific models, such as double layer LSTM and bidirectional LSTM, outperformed other models by giving low errors compared to other models. On the other hand, Transformer + LSTM and other models, such as ARIMA, offered worse results, with notably higher values. The same chart also reveals that LSTM-based models have a low standard deviation, denoting the relative stability and reliability of these models. The plot in Fig. 11 is a comprehensive illustration of the performance of various models across a set of evaluation metrics. The models include double layer LSTM, single-layer LSTM, GRU, Bidirectional LSTM, Transformer + LSTM, and ARIMA. Various performance indicators (MSE, RMSE, MAE, and others) can be placed on the horizontal axis, and their values are plotted on the vertical axis. This is evidenced by the fact that the lines are closely clustered for all the plots, except the x-axis, for such measures as RMSE, R2, and WI. The measured model performances demonstrate shared similarities between them. RRMSE and MSE assessment parameters show minor variations between the different regression models. Each colored line in Fig. 9 shows different models performing their assessment. Among all models, ARIMA exhibits specific characteristics specific to performance measurement parameters. This analysis confirms the overall identical performance of the models yet shows the necessity to focus on specific metrics for improving model efficiency. The radar Fig. 12 is a comprehensive tool that provides a detailed overview of the model's performance across various tests and measurements. Each axis on the radar plot corresponds to a specific metric: MSE, RMSE, MAE, and MBE, which are global measures of the error and bias of the problems. r, R2: Quantitative data on regularity and correspondence: coefficients of correlation and fits. This comprehensive tool equips the audience with a powerful means to evaluate model performance. RRMSE, NSE, WI: Measures that give information about the performance of models in terms of their accuracy and the resources used in their

Fig. 10 Comparison of models by mean and standard deviation



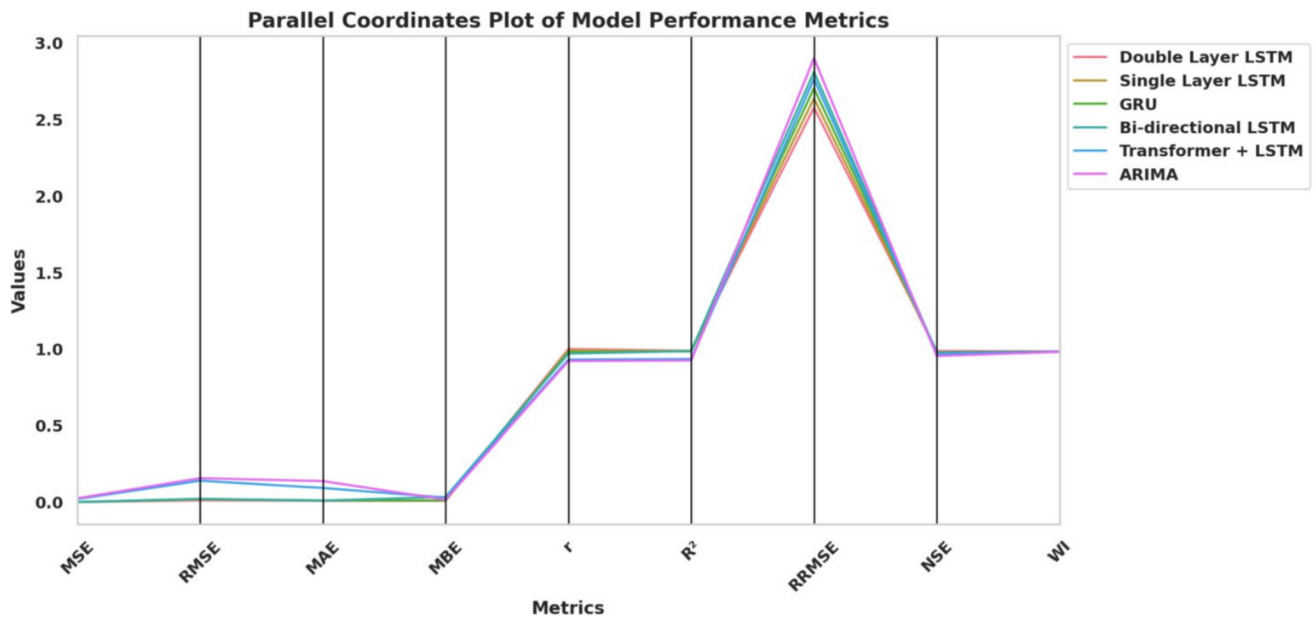
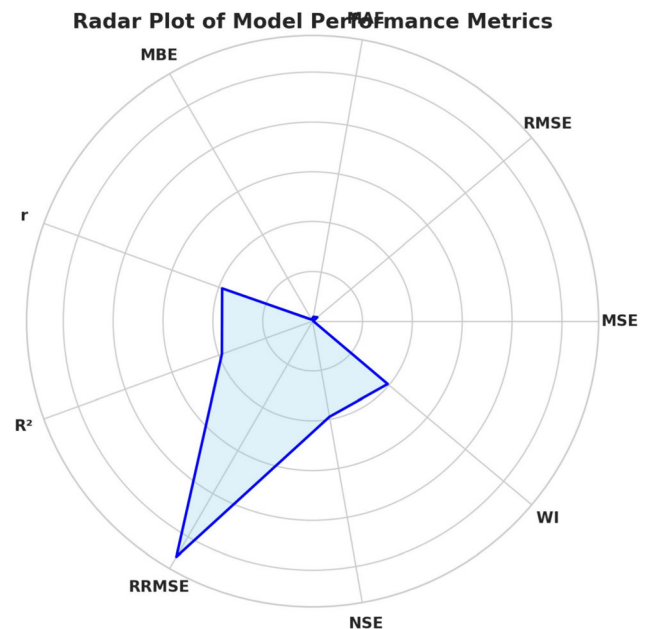


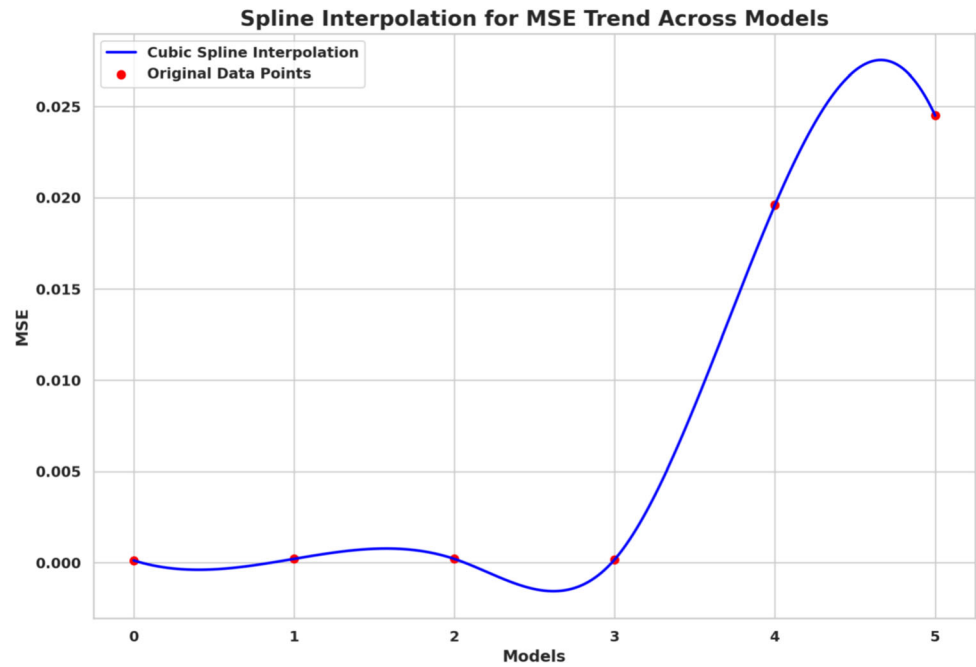
Fig. 11 Plotting parallel coordinates

Fig. 12 Radar plot of model performance metrics



creation. The blue line shows the connection of values of each measured parameter, making it a polygonal line. Higher values are located farther from the origin, making Employee metrics look better (if not error-based metrics such as MSE or RMSE, where lower values indicate better performance). The horizontal blue line covers the entire model performance area, also shaded in the Figure. Radar plots can also be used to flag the strengths and weaknesses of metrics together at a single glance. For example, a model might exhibit high R² while, at the same time, having high total error, as can be seen from the plot of the data. This Fig. 13 presents a visually engaging tool for understanding how the MSE varies across different models using Cubic Spline Interpolation. Raw data points are displayed in red dots, indicating the diagram's MSE values for particular models. These MSE values become accessible through the blue curved path that cubic spline interpolation produced. The independent variable exists along the X-axis as different models, while the dependent variable stays on the Y-axis as MSE

Fig. 13 Spline interpolation for MSE trend across models



Mixed Plot: Density + KDE for Metrics

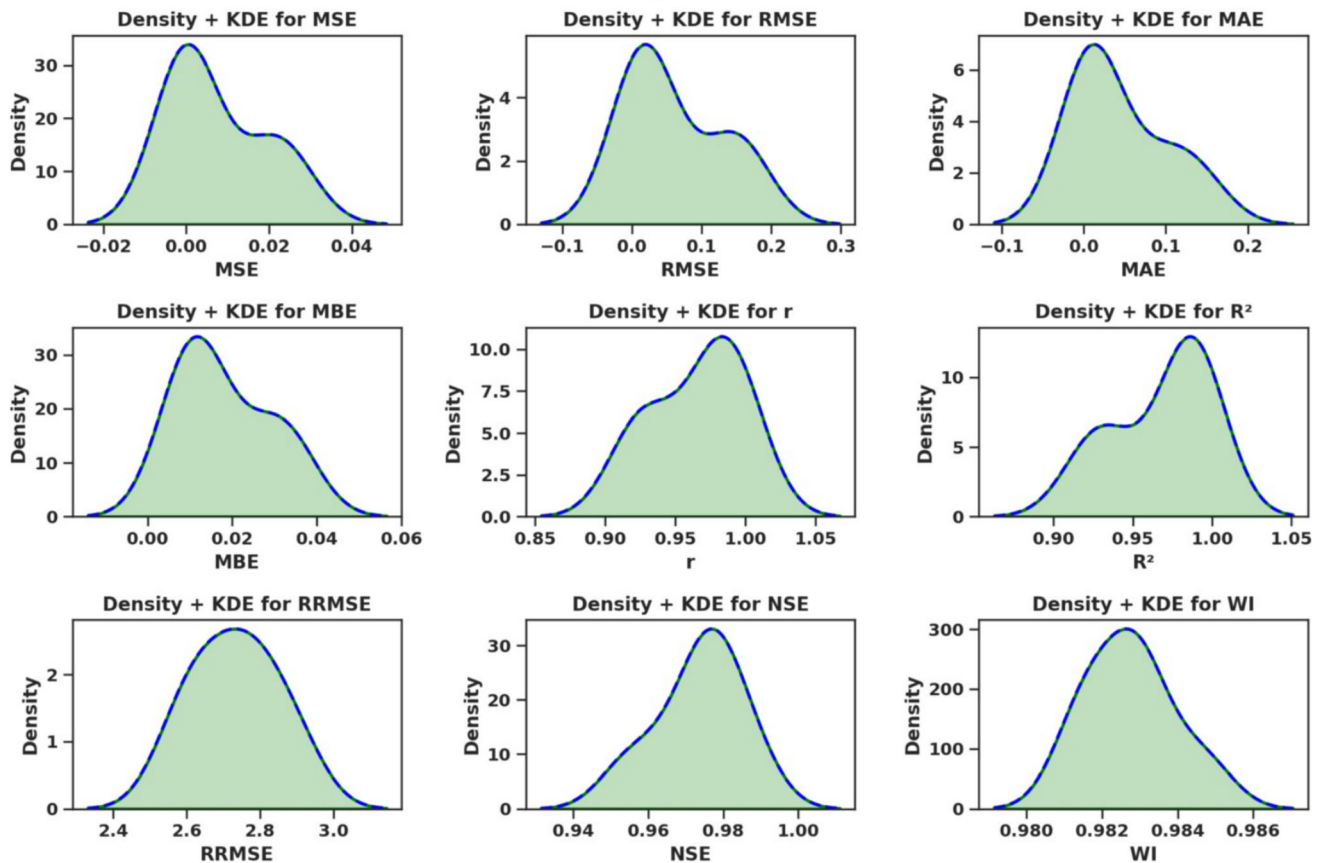
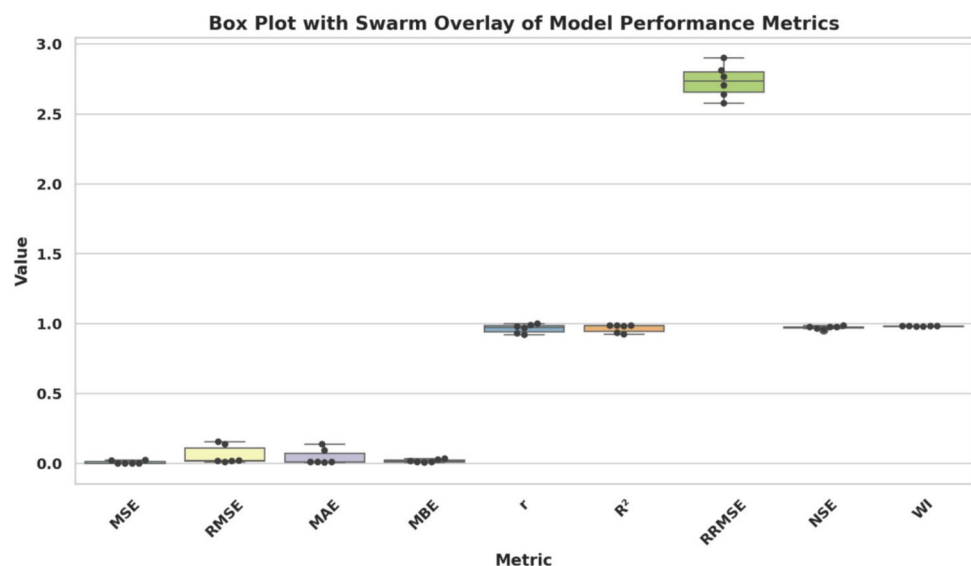


Fig. 14 Mixed Plot – Density + KDE

values. Cubic spline interpolation offers a visually pleasing way to understand the overall pattern of the set through its representation of error increases or decreases. For instance, we observe higher MSE for more significant model indices, indicating a change in model behavior. Figure 14 shows cases nine images illustrating the model prediction metrics through density plots and the Kernel Density Estimation technique. The KDE plot helps observers understand how the model operates across all evaluated metrics. MSE, RMSE, MAE: These are some of the ways most often used for evaluating the size of prediction errors. Mean Bias Error indicates that there is bias in the predictive computation. A positive value indicates a tendency to over-predict, while a negative value indicates under-predicting. r and R^2 : Employing such measures explains the relationship between the anticipated and actual case incidences. The R^2 part shows how much of the actual value variation can be attributed to the internal model. Root Mean Squared Error then scales the prediction errors to the extent that they can be compared more conveniently. Nash Sutcliffe Efficiency measures how well the model is. Such values signify that higher values are associated with better accuracy. A comparison of the actual and predicted values can be made by looking at the Willmott Index. Each plot has an area shaded in green to point at the metric values, and above the KDE plot, there is a smooth blue curve to accentuate density fluctuations. Performance figures are well conveyed by this form of visualization. The current Fig. 15 displays the box plot with swarm overlay showing the performance of the predictive models based on several evaluation metrics. This paper used MSE, RMSE, MAE, MBE, r , R^2 , RRMSE, NSE and WI to analyze the accuracy of the models. The box plot again helps to give a view of the data distribution for each of the metrics and shows the median, the range between the first and third quartiles and any outliers. The swarm overlay is depicted in the form of individual dots in black to provide an open for the spread and density of values as depicted by individual dots. However, the R^2 and WI diagnostics show a higher coefficient, highlighting the model's efficiency regarding variance explanation and comparing with real-world data. Hence, the model is evaluated based on MSE, RMSE and MAE, which reveal comparatively small figures indicating minimal prediction errors. In total, the graph of the outcoming metrics underlines the stability of the predictive model, as such characteristics are stable regardless of the measure chosen. Thus, high values of the correlation coefficients (r and R^2) along with low error measures support that the proposed model performs the outcome prediction effectively. Moreover, it is possible to note that most distributions are compact, indicating stable and relatively reliable performance in most tested conditions. Overall, this visualization does a great job of revealing the strengths and weaknesses of the model and informs a general audience of the kind of performance the model can deliver.

Fig. 15 Box Plot for performance of predictive models



7 Conclusion

Energy management system quality has emerged as a primary area of research in renewable energy because it stands as an essential distribution method for electricity. Wind speed predictions constitute a fundamental component of the system. The proposed method uses X-double LSTM technology to produce reliable wind speed forecasting results under differing environmental situations. When utilized in the selected dataset, the statistical model presented effectiveness in forecasting wind power. Multiple metrics, including MSE, RMSE, MAE, MBE, r , R^2 , RRMSE, NSE, and WI, show that the model achieves accurate results. The results prove that the proposed methodology effectively generates wind power forecasts. The implemented model uses an XAI-based shap algorithm for result explanation to improve transparency alongside predictive output interpretability. This algorithm identifies the most critical wind power-generating features to measure output levels, resulting in trustworthy model predictions. Combining X-double LSTM with an XAI-based shap enhances wind power generation prediction accuracy through an improved explanation of prediction results. Results from this work establish fundamental knowledge for deploying renewable energy technologies, especially wind power generation facilities. Further evaluation with extended datasets should occur to establish the proposed methodology's superior performance and enhanced prediction methods during future implementation.

Funding Open access funding provided by The Science, Technology & Innovation Funding Authority (STDF) in cooperation with The Egyptian Knowledge Bank (EKB). Not applicable.

Data availability The data that support the findings of this study are openly available at [<https://www.kaggle.com/datasets/jorgesandoval/wind-power-generation>].

Declarations

Conflicts of interest The authors declare that they have no conflicts of interest to report regarding the present study.

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